

What are JSON and Parquet Files?

Modern data pipelines often ingest data in **semi-structured or columnar formats** instead of traditional CSV files.

Two of the most common formats are:

Format	Type	Usage
JSON	Semi-structured	APIs, logs, application data
Parquet	Columnar structured	Big data analytics

JSON Files

JSON (**JavaScript Object Notation**) stores data in **key-value format**.

Example JSON file

```
{
  "customer_id":101,
  "name":"Rohith",
  "city":"Bangalore",
  "orders":[
    {"order_id":1,"amount":500},
    {"order_id":2,"amount":700}
  ]
}
```

Characteristics:

- Semi-structured
 - Supports nested data
 - Common in APIs and logs
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Parquet Files

Parquet is a **columnar storage format** optimized for analytics.

Example Parquet structure

order_id	customer_id	amount
1	101	500
2	102	900

Characteristics:

- Columnar storage
- Highly compressed
- Faster query performance

Parquet is commonly used in:

- Spark pipelines
- Data lakes
- Big data processing

Real-Time Project Scenario

Industry: **E-commerce Platform**

Data sources:

Source	File Format
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Customer API	JSON
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Transaction Data	Parquet
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Pipeline architecture

Application APIs



AWS S3 Data Lake



Snowflake External Stage



Raw Tables



Transformations



Analytics Tables

Step 1 – Create Storage Integration

To access cloud storage securely.

Example

```
CREATE STORAGE INTEGRATION s3_int
TYPE = EXTERNAL_STAGE
STORAGE_PROVIDER = S3
ENABLED = TRUE;
```

This allows Snowflake to access **S3 bucket data**.

Step 2 – Create Stage

Stage connects Snowflake to storage location.

Example

```
CREATE STAGE data_stage
URL='s3://company-data/'
STORAGE_INTEGRATION = s3_int;
```

Files stored in S3 can now be accessed.

JSON File Loading

Step 3 – Create JSON File Format

```
CREATE FILE FORMAT json_format
TYPE = JSON;
```

File formats help Snowflake **understand the file structure**.

Step 4 – Create Raw Table

JSON data is usually stored in **VARIANT** column.

Example

```
CREATE TABLE raw_customers
(
data VARIANT
);
```

Step 5 – Load JSON File

Example command

```
COPY INTO raw_customers
FROM @data_stage/customers/
FILE_FORMAT = json_format;
```

Snowflake automatically loads **JSON objects into the VARIANT column**.

Querying JSON Data

Example

```
SELECT
data:customer_id,
data:name,
data:city
FROM raw_customers;
```

Result

customer_id	name	city
101	Rohith	Bangalore

Handling Nested JSON

Example JSON

```
"orders":[
{"order_id":1,"amount":500},
{"order_id":2,"amount":700}
]
```

Extract nested data using **FLATTEN**.

Example

```
SELECT
data:customer_id,
order_item.value:order_id,
order_item.value:amount
FROM raw_customers,
LATERAL FLATTEN(input => data:orders) order_item;
```

Result

customer_id	order_id	amount
101	1	500
101	2	700

Parquet File Loading

Parquet is **already structured**, so Snowflake can load it directly into tables.

Step 1 – Create Parquet File Format

```
CREATE FILE FORMAT parquet_format  
TYPE = PARQUET;
```

Step 2 – Create Target Table

Example

```
CREATE TABLE sales_data  
(  
  order_id INT,  
  customer_id INT,  
  amount NUMBER  
);
```

Step 3 – Load Parquet Data

Example command

```
COPY INTO sales_data  
FROM @data_stage/sales/  
FILE_FORMAT = parquet_format;
```

Snowflake automatically **maps Parquet columns to table columns**.

Automatic Schema Detection

Snowflake can also load Parquet files **without predefined columns**.

Example

```
SELECT *  
FROM @data_stage/sales/  
(FILE_FORMAT => parquet_format);
```

This helps **preview data before loading**.

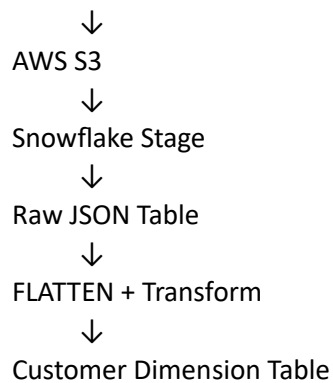
JSON vs Parquet Comparison

Feature	JSON	Parquet
Type	Semi-structured	Columnar structured
Storage	Larger size	Highly compressed
Performance	Slower	Faster analytics
Use Case	APIs / logs	Big data pipelines

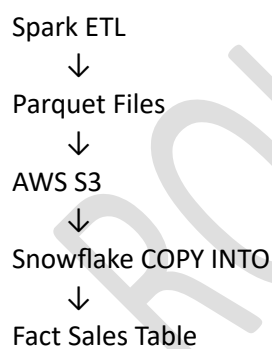
Real-Time Production Pipeline

Example real pipeline

Customer API (JSON)



Transaction pipeline



Best Practices

Production pipelines follow these rules:

- Store raw JSON in **VARIANT** columns
 - Use **FLATTEN** for nested arrays
 - Use Parquet for **large analytical datasets**
 - Maintain **staging tables before transformation**
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Common Interview Questions

Which Snowflake data type stores JSON data?

VARIANT

Which function extracts nested JSON arrays?

FLATTEN

Which file format is best for large analytical datasets?

Parquet

What command loads files into Snowflake tables?

COPY INTO

Quick Summary

Modern data pipelines use formats like **JSON and Parquet** to efficiently store and process data.

Snowflake supports both formats natively:

- JSON → stored using **VARIANT columns**
- Parquet → loaded directly into structured tables

This enables organizations to build **scalable and flexible data pipelines** for analytics.